Measuring the Effects of White Noise and Music on Anxiety Level Using a Commerical EEG

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Anxiety and anxiety disorders are one of the most prevalent mental health conditions, predominately affecting youth and young adults, ages 10-24 [1]. In the past 35 years, anxiety disorders have increased by 52% among adolescents and young adults. Daily anxiety affects nearly all parts of life, and is especially disruptive when under a high cognitive load. A 2007 study examined how anxiety impairs efficient cognitive functioning by taking attention away from the task at hand and instead focusing on the source of the anxiety [2]. However, the overall quality of performance does not neccessarily suffer, depending on compensatory stragegies used, like intentionally mentally working harder than neccesary. One strategy that requires less intentional mental application is playing white noise: a 2025 study examined how the anxiety levels of college students undergoing Cognitive Behavioral Therapy (CBT) were changed by white noise [3]. This study focuses on the difference in anxiety, depression, and sleep quality when only undergoing CBT, and while also listening to white noise in therapy. CBT represents a different dynamic to test-taking, but shares some qualities, such as therapy being an innate stessor, which can compound with external stressors (i.e.: needing to set boundaries with family). Anxiety levels in test-taking can be be monitored for changes from external factors like white noise or music using an Electroencephalogram (EEG) to measure changes in brain activity which is correlated to anxiety.

I. Literature Review

A. White Noise and Cognitive Load

A 2022 paper examines the effect of white noise on anxiety in operating room staff [4]. Operating rooms share many similarities with test-taking, presenting both inherent stressors (successfully operating on another human) and external stressors (an expectation for the staff to do their jobs well and quickly). Anxiety reduces performance at work, so any simple solution that could potentially decrease anxiety would represent a painless way to improve performance. Previous studies found that music significantly reduces stress, but also reduces the surgeon's auditory processing [5, p. 2]. A potential middle ground that addresses this issue are less audible, muffled sounds that draw less attention than music would. Thus, the study chose white noise from a suction machine, which is already commonly used in operating rooms.

The paper tests this hypothesis with a single-blind, crossover clinical trial on staff conducting cesarean sections, both with and without white noise from a suction machine. To test for differences in anxiety, the study measured hemodynamic parameters for the physical signs of anxiety, and used the Spielberger State-Trait Anxiety Inventory to measure anxiety level [4, pg. 1]. However, the study found no significant difference in anxiety whether the suction machine was running or not. The paper theorized that meaningless sounds like white noise are adapted to in the same way as other operating room noises, reducing its impact [4, pg. 3]. While these findings are inconclusive, there may be nuance lost in the somewhat chaotic setting of an operating room (chaotic in terms of multiple people talking, multiple sound sources, and a tangible source of high anxiety).

When isolating study participants with fewer uncontrollable variables, white noise was found to either notably increase sensory overload or to act as a sedative on the brain, likely depending on each individual's mental state and white noise intensity. In the former scenario, participants were tasked with folding a piece of paper into various configurations and difficulties, while introducing up to eight different conditions to mentally exhaust the participant [6]. The study found that sensory overload generally increased as more conditions were introduced, with white noise having the most signficant impact. However, the assesed level of sensory overload did not effect the participant's performance on the tasks, only their mental state. Alternatively, white noise can have a sedative effect on the prefrontal cortex, which presents as a calming effect [7]. This study differs from the previous two in that the participants were tested in calm, consistent environments, with no test or goal during the experiment aside from sitting in silence or listening to white noise. This format allowed for specific analysis of how white noise causes mental changes, but these positive effects may not be seen under high cognitive loads and anxiety levels.

B. Methods of Measuring Anxiety with an EEG

Commerical EEGs have high potential for lowering the bar of entry for those interested in studying brainwaves, at the cost of less accurate results. SingLEM is a pre-trained deep learning model trained on 357,000 hours of single-channel commercial EEG data [8]. Pre-training a deep learning model on such an expansive number of participants and hours of data allows it to differentiate between the brain signals and noise from blinking or moving. To validate SingLEM, it was tested as a fixed feature extractor; the model was not fine-tuned, and tested in identifying the correct limb in motor imagry tasks and mental states during cognitive tasks. Compared against its predicessors, SingLEM showed a 5-10% improvement across all tasks.

Alternatively, the raw electrical signal collected by an EEG can be converted into brainwaves using a Fourier Transformation. A 2017 study found that low Beta waves (12-17Hz) predict the participants percieved stress 94% of the time, while high Beta waves (18-30Hz) did not predict stress [9]. Filtering only for Beta waves and using a Naive Bayes machine allowed for accurate stress prediction with low computational cost.

EEGs with multiple sensors have more potential methods of data analysis. Thanks to the sensors being placed in multiple locations, a model could track the fluctuations in each observed feature to account for the distortion inherent to reading brainwaves through the skull. A 2023 study implemented this using an EEG with 14 channels, and tracked Hjorth parameters to measure the properties of the signal over time, sub-band powers like Alpha and Beta waves, and an Engagement Index, a calculated ratio Alpha, Beta, and Theta waves [10]. Tracking the changes in these features across the skull allowed the team to generate normalized distributions, allowing for more accurate data collection.

II. Questions and Hypotheses

The above examined studies on axiety and EEG anxiety assessment provides a solid base to extrapolate how noise would effect anxiety while test-taking. White noise in the operating room showed no difference in anxiety level, while music decreased both anxiety and auditory processing [4], [5]. In an operating room, auditory processing is critical, but in many other stressful situations, it is much less important. In a more controlled scenario, it may be more likely that the white noise would calm the participant [7].

While performing controlled tasks, what is the effect of white noise on task anxiety? It is likely that listening to white noise shows a statistically significant reduction in anxiety and no change in performance [7]. Though unexpected, anxiety levels may remain the same or rise while listening to white noise and compared to the task done in silence [4],

[6]. Through collecting data using a commercial EEG, we will test the effect of white noise on anxiety-inducing tasks. Additionally, we will repeat the same tasks while listening to music with and without lyrics, and expect no change in performance but a decrease in anxiety [11].

Using a wearable EEG that measures stress, we will test these claims by repeatedly taking tests under various auditory conditions. EEGs have already proven to be accurate in tracking stress, and allow for continuous monitoring throughout test-taking [8]. With an EEG, we can track how anxiety levels shift throughout the test-taking process, allowing for insights like when axiety is at its highest, and how much various kinds of noise dampen that peak. To interpret this data, we will fine-tune SingLEM on our collected data, and perform brainwave band analysis, focusing on the Beta band [8], [9].

III. Device Design

To realize a wearable anxiety monitor, a NeuroSky MindWave Mobile 2 is being used. This headset was released in 2018, and included first-party software support on Windows, Android, and iOS [12]. To track data, we will use NeuroSky's eegID Android app, which saves all collected data as a CSV [13]. While there are other alternatives, including connecting the headset directly to a computer using a python script or program, the mobile app is sufficient, with all data analysis happening afterwards. Additionally, because eegID has not recieved updates since late 2015, it will be easier to pair it to an older phone - in this case, a Motorola Moto X4, which was released in 2017.

Fig. 1: Hardware data flow and synchronization architecture.

IV. Methods Plan

To measure anxiety, we will perform two tasks in silence, and while listening to white noise, lyrical music, and lyric-less music. For the purpose of training SingLEM, we will also take data while aware but unstressed. Based loosely off of Peplau's definitions of levels of anxiety, we chose Level 1 to be a cognitively aware baseline, and Level 2 as light anxiety while under cognitive load, with our Level 2 corresponding to Peplau's Level 1 [14]. To collect Level 1 data, we chose two tasks: reading a book, and listening to music. For Level 2 data, we chose a modified Stroop test and a Typing test.

The Stroop test traditionally is formatted as a list of words in various colors, sometimes with the words being the names of those colors [15]. The participant must name the color of the word, not the word itself. To avoid the additional movement introduced by speaking, we will instead play the Stroop game using a Python script, with four colors mapped to keys on the keyboard. This game is cognitively challenging by forcing the brain to simultaneously process the word and the color of the word, with the color detection being the much less practiced skill of the two. Additionally, we will implement a correct answer counter, adding inherent pressure as the streak increases [16].

Our Typing test is similar to most online typing games, with a feed of words to type as fast as possible. Like the Stroop test, this game will implement a correct word counter to increase anxiety during the test [16].

A. Collecting Data

We will use a NeuroSky Mindwave Mobile 2 to collect data and follow the included instructions for positioning: the sensor over the left eyebrow and ground clip on the left ear [12]. When initially putting on the headset, we will readjust it until the 'Poor Signal' field drops to 0, and quickly returns there after blinking. We then set the recording limit to '5 minutes', and the recording interval to 'All'. All data will be collected in the same room, sitting at a desk in front of a computer. When the computer is not being used, it will be off. If the Stroop or Typing test is being taken, we will start the EEG recording first, then start the game on the computer. If white noise or music is needed, it will be played through wireless earbuds.

TABLE I: eegID Telemetry Fields and Correlated Brainwave States [12], [13]

eegID Field	Frequency Range	Description and Meaning
PoorSignal	N/A	A hardware metric indicating the quality of the sensor contact and signal integrity.
EEG Raw Value	N/A	Unfiltered analog values captured directly from the EEG sensor.
EEG Raw Value Volts	N/A	The raw EEG data scaled and converted into standard voltage units.
Beta Low	13–17 Hz	Reflects a state of being relaxed yet focused, integrated, and actively thinking while remaining aware of surroundings.
Beta High	18–30 Hz	Indicates high alertness and mental activity (e.g., math, planning), and general activation of mind/body functions, but can also reflect agitation.

B. Methods of Measuring Anxiety

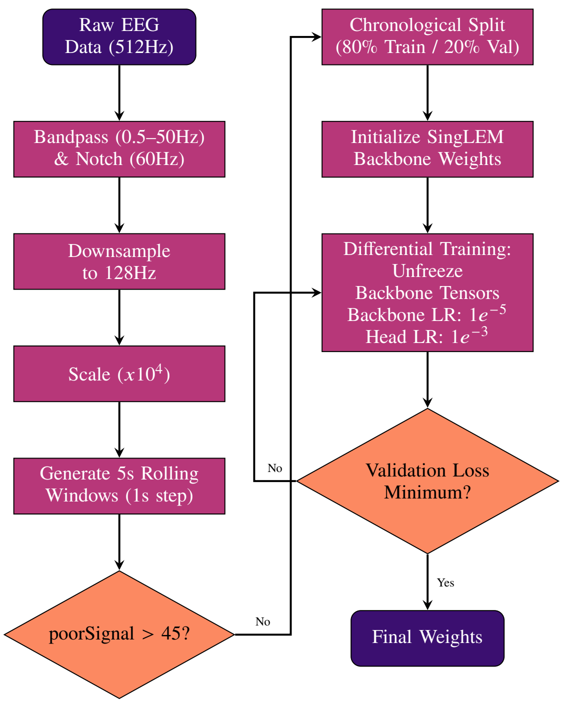

Fig. 2: System Architecture: EEG preprocessing, artifact rejection, and SingLEM fine-tuning with automated Early Stopping.

SingLEM is fine-tuned using the silent level 1 and level 2 data. We are basing our fine-tune on the example included in SingLEM's GitHub [17]. First, our data is passed through multiple filters to eliminate external electrical noise, data points with head connection, and downsampled to match how SingLEM was trained [18]. The first 60 seconds of all datasets are removed to account for any initial movement, and to wait until the anxiety from the Level 2 tasks have begun to build up. Lastly, the model is partially unfrozen to compensate for the small size of the datapool, with a learning rate of $(1e^{-}5)$.

This format fine-tunes SingLEM into a binary classifier, which can then be used to classify our data under various auditory conditions and compare to the silent data. To perform further data analysis, the Expected Value will give detailed information on how close to Level 1 or 2 the participant is over time. Calculating the Expected Value avoids the sometimes inaccurate outputs from only using a softmax.

$$E[L] = (P_1 \times V_1) + (P_2 \times V_2) \tag{1}$$

Where:

- $E[L]$ is the Expected Level (continuous cognitive load score).
- P_1 is the Softmax probability assigned to Class 1.
- V_1 is the numerical value assigned to Level 1 ($V_1 = 1.0$).
- P_2 is the Softmax probability assigned to Class 2.
- V_2 is the numerical value assigned to Level 2 ($V_2 = 2.0$).

Additionally, we are implementing the Beta band classification from Saeed's paper (PSS) [9]. We can skip Saeed's preprocessing steps, since our EEG already outputs low and high Beta waves. In the paper, they related Beta waves to a Percieved Stress Score (PSS); in our analsis we will compare the result to our labelled data.

$$PSS = b_0 + b_1\beta_L + b_2\beta_H \tag{2}$$

Where:

- PSS is the predicted Perceived Stress Scale score[cite: 135].
- b_0 is the intercept ($b_0 = -2.35$)[cite: 135, 160].
- b_1 is the regression weight for the low beta wave ($b_1 = 2.29 \times 10^{-6}$)[cite: 135, 160].
- β_L is the value of the low beta wave (13-17 Hz)[cite: 128, 135].
- b_2 is the regression weight for the high beta wave ($b_2 = 1.20 \times 10^{-7}$)[cite: 135, 160].
- β_H is the value of the high beta wave (18-30 Hz)[cite: 128, 135].

Finally, the results from SingLEM and the PSS equation are checked for normality, and evaulated using either a One-Way ANOVA (for normal and equally distributed data) or a Kruskal-Wallis test (for non-parametric data).

V. Analysis

Upon fine-tuning SingLEM, we got a validation accuracy of 73.7% after six epochs [18]. This top accuracy was reached after significant difficulties with over- and under-fitting to the dataset. While this result does not appear to suffer from either of those issues, it is not clear that this is a good fine-tune.

We evaluated the statistical significance of four paramters across two tasks: performance throughput (keys or words per second), performance accuracy (how mnay mistakes were made), EEG calculated anxiety using PSS, and SingLEM predicted anxiety [18]. To test for normality and variance, all parameters first passed through a Shaprio-Wilk test and a Levene's test. Of these eight, all used One-Way ANOVA except for performance throughput for both the Stroop and Typing tests.

For the Stroop task, every parameter returned the null result, with p-values well above the 0.05 threshold. For the Typing task, accuracy, PSS, and predicted anxiety were not statistically significant, but performance throughput did have a significant difference (p=0.0379) under the Kruskal-Wallis test. Aside from the average number of words per second, none of the measured performance or anxiety data contained a significant difference between the silent data and the white noise, lyrical, or lyric-less data.

VI. Results and Conclusion

Unlike the papers cited examining the effect of white noise, we were not able to find a significant difference in anxiety level between auditory levels. Instead, the Stroop or Typing task appeared to override the environmental audio. The one paramter that did show a significant difference, the Typing task's throughput, may instead point towards the background audio being a motor pacer, speeding up or slowing down the speed of typing. Lastly, the anxiety levels calculated between the SingLEM fine-tune and the PSS equation were similar. This indicates that the SingLEM fine-tune was likely accurate despite the lower than hoped accuracy score of 73.7%.

Our results not lining up with our initial hypothesis, and the previous papers examing the effect of white noise, is likely due to flaws in our methodology. While the Stroop and Typing tasks are good measures of anxiety under cognitive load, the amount of data required to properly fine-tune may have eliminated the anxiety-inducing elements of the tasks. Additionally, the readings were taking over multiple sessions, meaning that the headset was taken on and off multiple times, resulting in small differences in sensor placement on the forehead. Theoretically, SingLEM should be pre-trained thoroughly enough to ignore these inconsistencies, but it likely degraded the quality of our data.

Going forward, the data collection phase should shift away from gathering 15-60 minutes of data on each condition to gathering less data on more tasks. The main difficulty with this approach would be designing tasks without too much physical movement, which could introduce too much noise to give meaningful results.

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TABLE II: Behavioral Performance Metrics by Task and Auditory Condition (Participant: Andy)

Task	Condition	Runs	Avg Inputs	Avg Errors	Throughput (/sec)	Accuracy (%)
Stroop	Silent	6	458.5	26.8	1.54	94.15
	WhiteNoise	3	453.7	25.0	1.52	94.49
	Music	3	480.3	38.3	1.61	92.02
	MusicNL	3	531.7	31.0	1.78	94.17
Typing	Silent	6	241.5	54.5	0.81	81.59
	WhiteNoise	3	266.3	49.7	0.89	84.28
	Music	3	267.7	57.7	0.90	82.27
	MusicNL	3	268.7	50.0	0.90	84.31

TABLE III: Statistical Analysis of Performance and EEG Metrics (Stroop Task)

Metric Assessed	Normality	Variance	Statistical Test	p-value	Significant?
Performance Accuracy	Pass	Equal	One-Way ANOVA	0.6535	No
Performance Throughput	Pass	Unequal	Kruskal-Wallis	0.1149	No
EEG Calculated PSS	Pass	Equal	One-Way ANOVA	0.4603	No
EEG Predicted Anxiety	Pass	Equal	One-Way ANOVA	0.9643	No

TABLE IV: Statistical Analysis of Performance and EEG Metrics (Typing Task)

Metric Assessed	Normality	Variance	Statistical Test	p-value	Significant?
Performance Accuracy	Pass	Equal	One-Way ANOVA	0.0781	No
Performance Throughput	Pass	Unequal	Kruskal-Wallis	**0.0379**	**Yes**
EEG Calculated PSS	Pass	Equal	One-Way ANOVA	0.1642	No
EEG Predicted Anxiety	Pass	Equal	One-Way ANOVA	0.2914	No

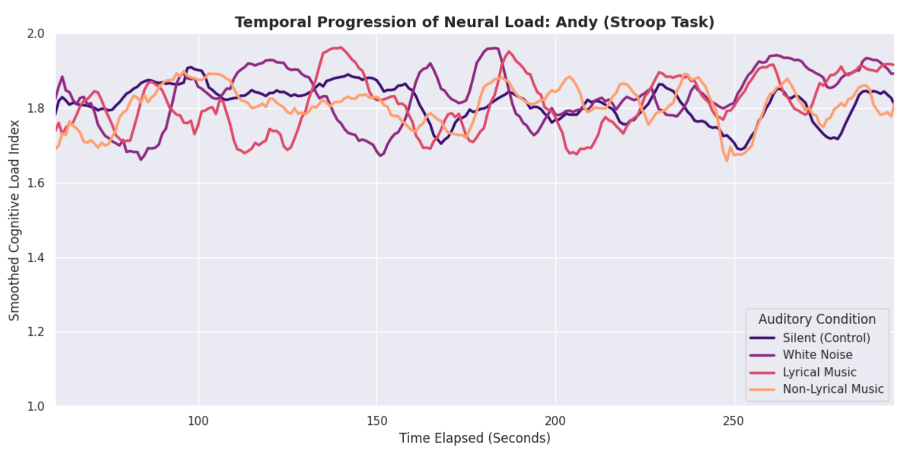

TABLE V: Anxiety over time during the Stroop Test.

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